

JavaScript Switch-Case Statement Assignment

Section 1: Basic Switch-Case with Strings

Question 1.1: What will be the output of the following code?

javascript

```
let browser = 'firefox';
```

```
switch (browser) {  
  case 'chrome':  
    console.log('launching chrome');  
    break;  
  case 'firefox':  
    console.log('launching firefox');  
    break;  
  case 'edge':  
    console.log('launching edge');  
    break;  
  default:  
    console.log('browser not supported');  
    break;  
}
```

Question 1.2: What will be the output?

javascript

```
let environment = 'staging';
```

```
switch (environment.toLowerCase()) {  
  case 'dev':  
    console.log('development environment');
```

```
        break;
    case 'staging':
        console.log('staging environment');
        break;
    case 'prod':
        console.log('production environment');
        break;
    default:
        console.log('unknown environment');
        break;
}
```

Question 1.3: What will be the output?

javascript

```
let testType = 'API';

switch (testType.trim().toLowerCase()) {
    case 'ui':
        console.log('running UI tests');
        break;
    case 'api':
        console.log('running API tests');
        break;
    case 'mobile':
        console.log('running Mobile tests');
        break;
    default:
        console.log('test type not recognized');
        break;
}
```

Section 2: Switch-Case with Numbers

Question 2.1: What will be the output?

javascript

```
let score= 85;
```

```
switch (score) {  
  case 100:  
    console.log('perfect score');  
    break;  
  case 90:  
    console.log('excellent');  
    break;  
  case 85:  
    console.log('very good');  
    break;  
  default:  
    console.log('keep improving');  
    break;  
}
```

Question 2.2: What will be the output?

javascript

```
let statusCode = 404;
```

```
switch (statusCode) {  
  case 200:  
    console.log('success');  
    break;
```

```
case 404:
    console.log('not found');
    break;
case 500:
    console.log('server error');
    break;
default:
    console.log('unknown status');
    break;
}
```

Section 3: Switch-Case with Boolean Values

Question 3.1: What will be the output?

javascript

```
let isVisible = true;
```

```
switch (isVisible) {
  case true:
    console.log('element is visible');
    break;
  case false:
    console.log('element is hidden');
    break;
  default:
    console.log('invalid state');
    break;
}
```

Question 3.2: What will be the output?

javascript

```
let testPassed = false;

switch (testPassed) {
  case true:
    console.log('test passed');
    break;
  case false:
    console.log('test failed');
    break;
  default:
    console.log('test result unknown');
    break;
}
```

Section 4: Default Case Placement

Question 4.1: What will be the output when default is not at the end?

```
javascript

let browser = 'opera';

switch (browser) {
  case 'chrome':
    console.log('chrome launched');
    break;
  case 'firefox':
    console.log('firefox launched');
    break;
  default:
    console.log('browser not in main list');
    break;
}
```

```
    case 'opera':  
        console.log('opera launched');  
        break;  
    case 'safari':  
        console.log('safari launched');  
        break;  
}
```

Question 4.2: What will be the output?

javascript

```
let platform = 'android';  
  
switch (platform) {  
    case 'windows':  
        console.log('windows platform');  
        break;  
    default:  
        console.log('other platform');  
        break;  
    case 'mac':  
        console.log('mac platform');  
        break;  
    case 'linux':  
        console.log('linux platform');  
        break;  
}
```

Section 5: Missing Break Statements (Fall-through)

Question 5.1: What will be the output when break statements are missing?

javascript

```
let grade = 'B';
```

```
switch (grade) {
```

```
  case 'A':
```

```
  }
```

```
    console.log('excellent');
```

```
  case 'B':
```

```
    console.log('good');
```

```
  case 'C':
```

```
    console.log('average');
```

```
  default:
```

```
    console.log('grade processed');
```

Question 5.2: What will be the output?

javascript

```
let priority='high';
```

```
switch (priority) {
```

```
  case 'critical':
```

```
    console.log('critical priority');
```

```
    break;
```

```
  case 'high':
```

```
    console.log('high priority');
```

```
  case 'medium':
```

```
    console.log('medium priority');
```

```
        break;
    case 'low':
        console.log('low priority');
        break;
}
```

Section 6: Real-world Automation Scenarios

Question 6.1: What will be the output for this locator strategy switch?

javascript

```
let locatorType = 'xpath';

switch (locatorType) {
  case 'id':
    console.log('finding element by ID');
    break;
  case 'xpath':
    console.log('finding element by XPath');
    break;
  case 'css':
    console.log('finding element by CSS');
    break;
  case 'text':
    console.log('finding element by text');
    break;
  default:
    console.log('invalid locator type');
    break;
}
```


Question 6.2: What will be the output for this user role scenario?

javascript

```
let userRole = 'admin';

switch (userRole) {
  case 'admin':
    console.log('full access granted');
    break;
  case 'customer':
    console.log('customer access granted');
    break;
  case 'guest':
    console.log('limited access granted');
    break;
  default:
    console.log('access denied');
    break;
}
```

Question 6.3: What will be the output for this payment method scenario?

javascript

```
let paymentMethod = 'UPI';

switch (paymentMethod.toLowerCase()) {
  case 'upi':
    console.log('processing UPI payment');
    break;
  case 'card':
    console.log('processing card payment');
```

```
        break;
    case 'paypal':
        console.log('processing PayPal payment');
        break;
    case 'cash':
        console.log('processing cash payment');
        break;
    default:
        console.log('payment method not supported');
        break;
}
```

Section 7: Multiple Cases for Same Action

Question 7.1: What will be the output?

javascript

```
let browser = 'edge';

switch (browser) {
    case 'chrome':
    case 'chromium':
        console.log('chromium-based browser');
        break;
    case 'firefox':
        console.log('mozilla browser');
        break;
    case 'edge':
    case 'ie':
        console.log('microsoft browser');
        break;
    default:
        console.log('other browser');
```

```
        break;
    }
}
```

Question 7.2: What will be the output?

javascript

```
let day = 'saturday';

switch (day.toLowerCase()) {
  case 'monday':
  case 'tuesday':
  case 'wednesday':
  case 'thursday':
  case 'friday':
    console.log('weekday');
    break;
  case 'saturday':
  case 'sunday':
    console.log('weekend');
    break;
  default:
    console.log('invalid day');
    break;
}
```

Section 8: Complex Switch-Case Scenarios

Question 8.1: What will be the output for this loan interest rate scenario?

javascript

```
let loanType = 'home';
```

```
switch (loanType) {  
  case 'home':  
    console.log('interest rate: 8%');  
    break;  
  case 'car':  
    console.log('interest rate: 10%');  
    break;  
  case 'education':  
    console.log('interest rate: 6%');  
    break;  
  case 'personal':  
    console.log('interest rate: 12%');  
    break;  
  
  }  
  default:  
    console.log('loan type not available');  
    break;  
}
```

Question 8.2: What will be the output for this ride booking scenario?

javascript

```
let rideType = 'sedan';
```

```
switch (rideType) {  
  case 'mini':  
    console.log('mini car booked - $5');  
    break;  
  case 'sedan':
```

```
        console.log('sedan car booked - $8');

        break;

    case 'suv':

        console.log('SUV booked - $12');

        break;

    case 'bike':

        console.log('bike booked - $3');

        break;

    default:

        console.log('ride type not available');

        break;

}
```

Section 9: Switch vs If-Else Comparison

Question 9.1: What will be the output of both approaches?

javascript

```
let testResult = 'pass';
```

// Switch approach

```
switch (testResult) {

    case 'pass':

        console.log('test passed - switch');

        break;

    case 'fail':

        console.log('test failed - switch');

        break;

    default:

        console.log('unknown result - switch');

        break;

}
```

// If-else approach

```
if (testResult === 'pass') {  
    console.log('test passed - if-else');  
} else if (testResult === 'fail') {  
    console.log('test failed - if-else');  
} else {  
  
}  
console.log('unknown result - if-else');
```

Question 9.2: What will be the output when no case matches?

javascript

```
let device = 'smartwatch';
```

```
switch (device) {  
    case 'mobile':  
        console.log('mobile device');  
        break;  
    case 'tablet':  
        console.log('tablet device');  
        break;  
    case 'desktop':  
        console.log('desktop device');  
        break;  
    default:  
        console.log('device type: ' + device);  
        break;  
}
```